HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Supported the development and maintenance of user needs and system requirements for active implantable medical devices under senior engineer guidance.
- Assisted in maintaining requirements traceability between user needs, design outputs, and verification/validation activities using established documentation protocols.
- Participated in system architecture and integration discussions, developing an understanding of how diverse system elements interact within complex medical devices.
- Supported risk management activities, including hazard analysis and risk assessments, adhering to medical device development processes and regulatory standards.
- Assisted with the planning and execution of system-level verification and validation activities, documenting test results and investigating technical issues.
- Helped prepare and maintain clear, accurate systems engineering documentation and design records, ensuring compliance with internal standards.
- Collaborated with cross-functional teams including Electrical, Software, Mechanical, Manufacturing, Quality, and Regulatory colleagues to support product development.

SKILLS

Systems Engineering – Requirements Development, System Architecture, Risk Management, Verification and Validation, Product Development, Design Controls

Medical Device Development – Active Implantable Devices, Medical Technology, Regulatory Compliance, Hazard Analysis

Software Engineering – Java, Spring Boot, Python, REST APIs, SQL, PostgreSQL, MySQL, Oracle

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git

Data Analytics & BI – Power BI, Tableau, R/RStudio, pandas, NumPy, scikit-learn, DAX, Power Query

Project Management & Tools – Agile Methodologies, Scrum, Jira, Confluence

PROJECTS

Medical Device Requirements Traceability System

- Developed a Python-based tool to automate the mapping and validation of requirements across user needs, design specifications, and verification test cases for a hypothetical implantable device.
- Integrated with a PostgreSQL database to store and query requirement relationships, enabling efficient impact analysis for proposed changes and ensuring adherence to design controls.
- Created interactive dashboards using Power BI to visualize traceability status and highlight potential gaps for review by project stakeholders, improving oversight.

System Integration Testing Framework for Embedded Software

- Designed and implemented a modular integration testing framework in Java for a simulated embedded system, focusing on interface validation and error handling.
- Authored test scripts to simulate interactions between software components and hardware interfaces, verifying system behavior under various operational conditions.
- Documented test procedures and results, contributing to the overall verification and validation evidence package for the product development lifecycle.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)